

# Exploratory Data Analysis in Excel

Uncovering patterns, trends, and outliers in sales data — a DecodeLabs internship project.

DATA ANALYTICS

MICROSOFT EXCEL

EDA



# What Is EDA?



## Exploratory Data Analysis

Before modeling, you **explore**. EDA means inspecting data to:

- **Understand structure**
- **Spot anomalies**
- **Reveal patterns**
- **Guide decisions**



# Tools Used in Excel



## Formulas

SUMIF, COUNTIF, ROUND, DATE



## Pivot Tables

Summarize & segment data



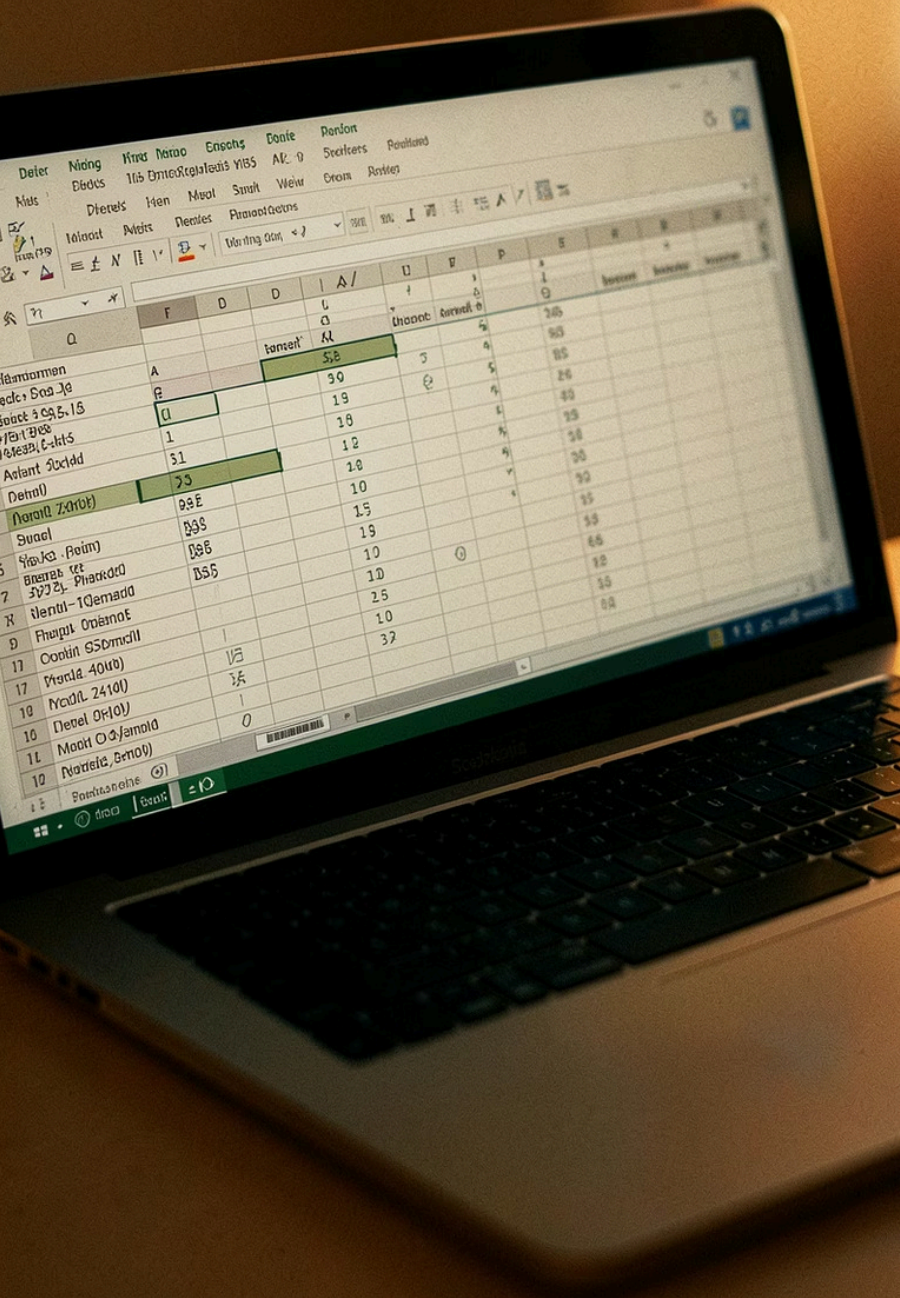
## Charts

Pie, bar, scatter, boxplot



## Data Cleaning

Remove duplicates, handle nulls



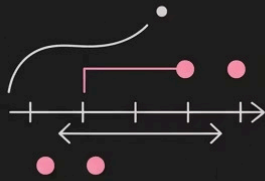
# Key Techniques Applied



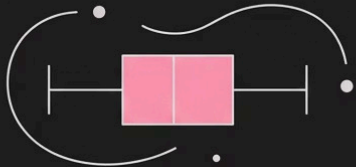
**MEAN &  
CENTRAL  
TENDENCY**



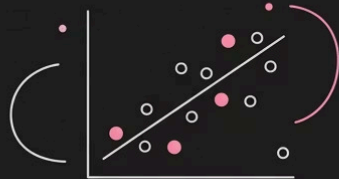
**COUNT &  
COUNTIF  
FREQUENCY**



**IQR METHOD  
OUTLIER  
DETECTION**



**BOXPLOT  
LOGIC  
DISTRIBUTION  
VISUAL**



**CORRELATION  
ANALYSIS  
VARIABLE  
RELATIONSHIPS**

## From Raw Data to Insight

Each technique answers a different question — together they build a complete picture of the dataset.

- ❏ **IQR Method:** Identifies outliers beyond  $1.5 \times$  the interquartile range.

# Key Findings at a Glance

**40%**

## Social Media Referrals

Of total orders — highest referral source

**≈30%**

## Cancellation Rate

Orders cancelled across the dataset

**R370**

## Average Unit Price

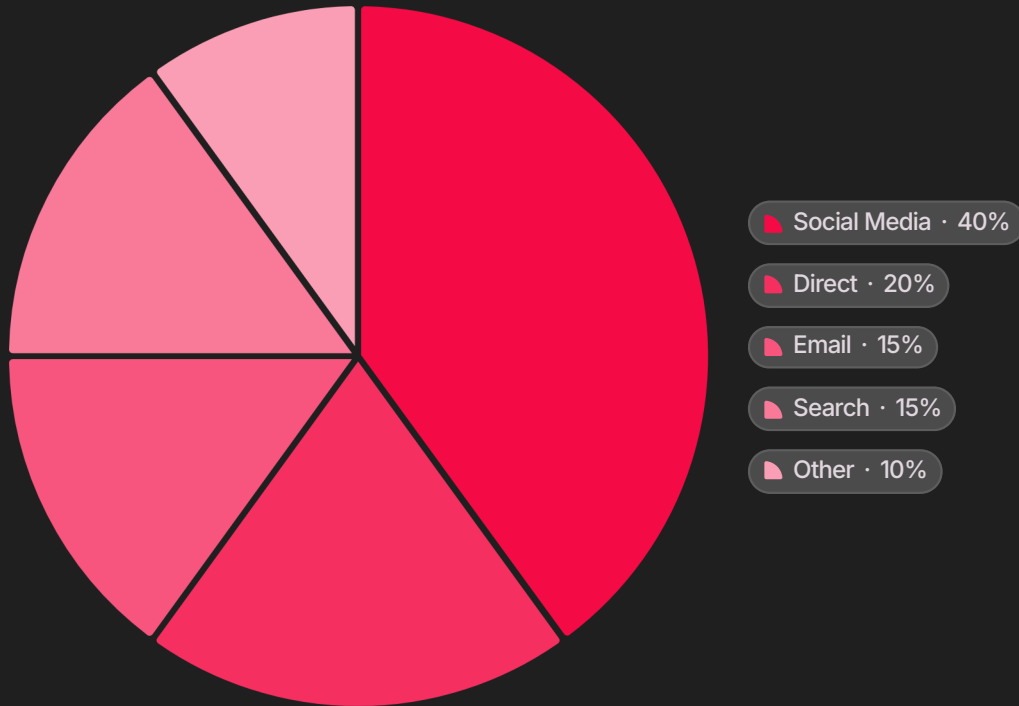
Mean price per item sold

**5**

## Charts Produced

Visual outputs from the analysis

# Referral Sources & Order Status



## Social Media Dominates

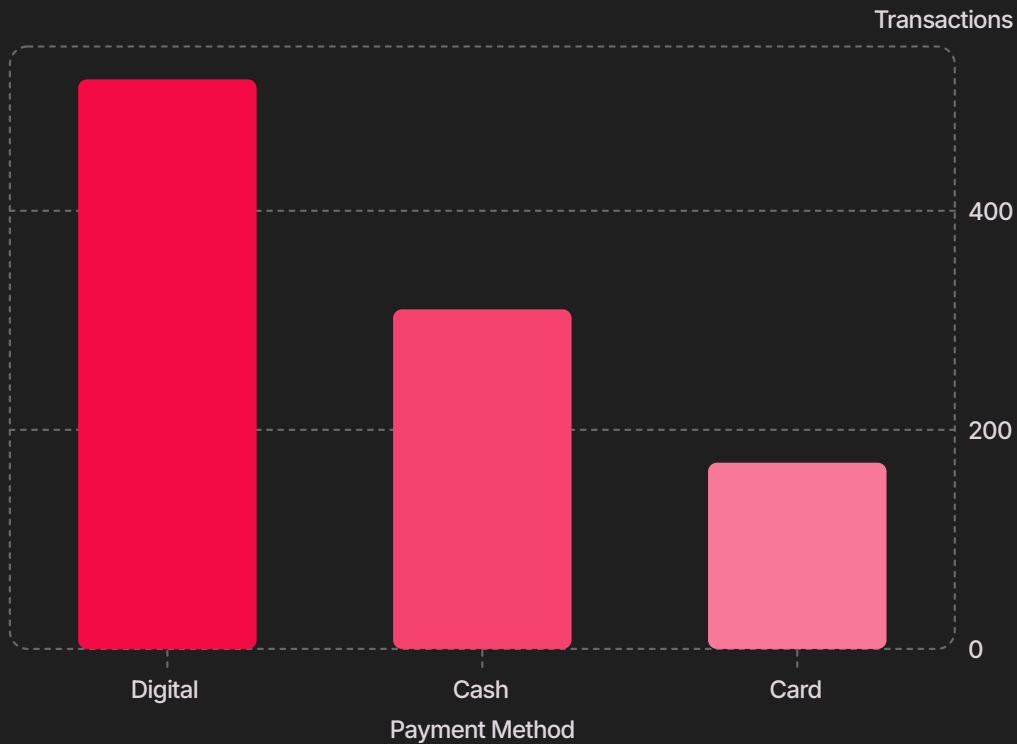
At **40%**, social media outpaces all other channels combined. Targeted campaigns here offer the highest ROI.

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## Order Status Breakdown

Shipped, cancelled, returned, and pending orders visualized via bar chart — **~30% cancellation rate** warrants investigation.

# Unit Price Distribution



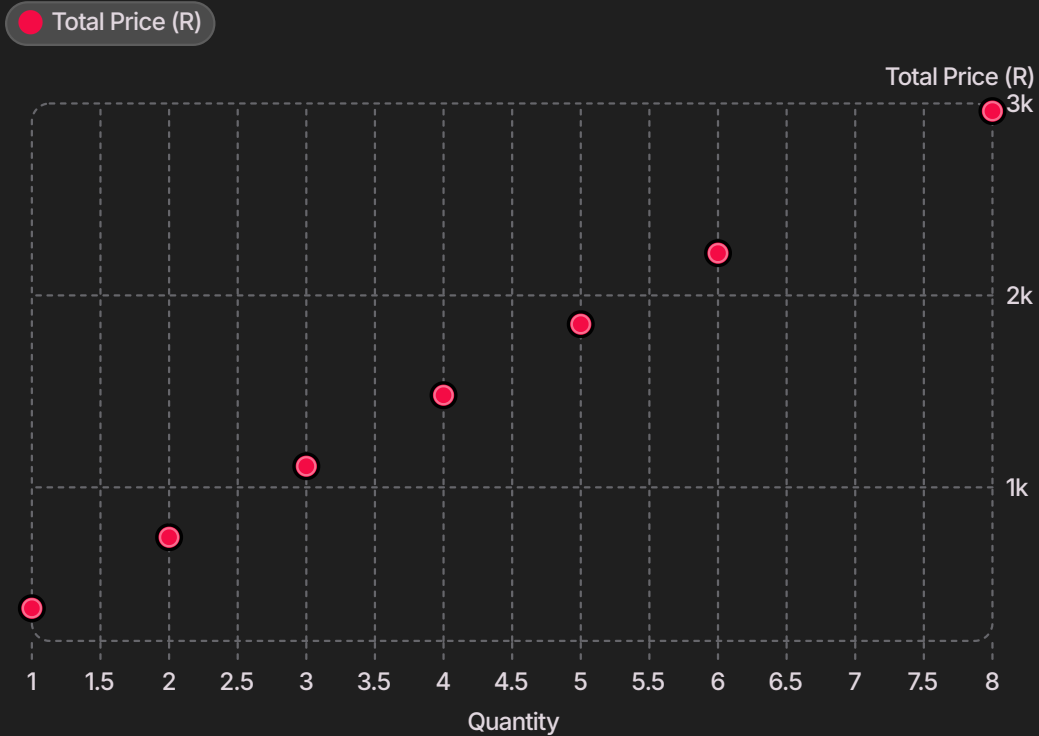
## Digital vs. Cash Trends

Digital payments lead — reflecting a shift toward cashless transactions in the dataset.

## Boxplot: Price Variability

Unit price boxplot highlights **outliers** and spread. IQR method flags extreme values for review.

# Correlation: Quantity vs. Total Price



## Positive Correlation Confirmed

As quantity increases, total price rises proportionally — validating data integrity and pricing logic.



**Why it matters:** A clean linear relationship confirms no systematic pricing errors in the dataset.



# Submission Checklist

## ✓ Cleaned Dataset

Cleaned\_Dataset.xlsx uploaded

## ✓ Project Summary

Word or PDF with analysis steps

## ✓ Chart Screenshots

All five visuals attached

## ✓ Formula Verification

SUMIF, COUNTIF, ROUND, DATE  
checked

## ✓ Cover Page

Name and batch details included

# The "So What?" Test

## DecodeLabs Values

Clarity

Originality

Insight

## Frame Every Finding

"Social media referrals account for 40% of total orders — suggesting targeted Instagram campaigns could further boost engagement."

Don't just report numbers. Explain **why they matter** and what action to take.